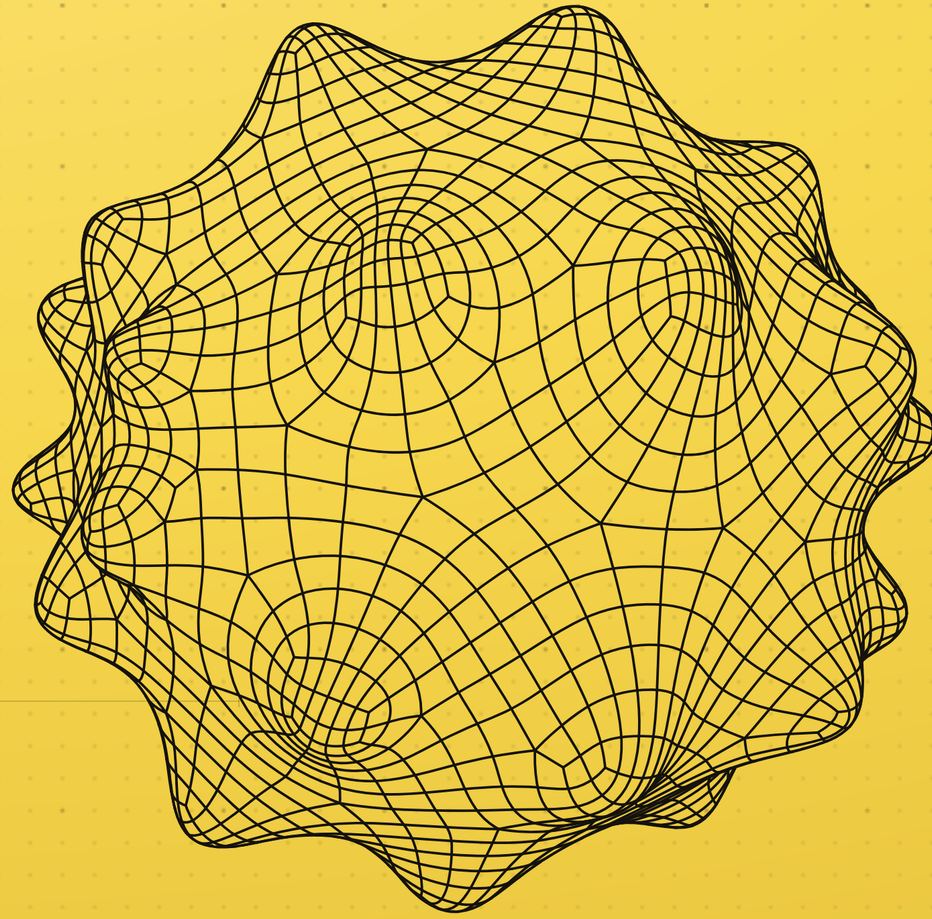
INTERNATIONAL CONFERENCE ON AGRICULTURE-CENTRIC COMPUTATION

AgriPerceiver

*A parameter-efficient vision–language model for **structured** macroscopic crop phenotyping*



Vatsal Khanna* · **Samridhi Saini*** · **Davinder Singh*** · **Kumkum Kwatra**

DR B R AMBEDKAR NIT JALANDHAR · ANNAM.AI, IIT ROPAR · * EQUAL CONTRIBUTION

◆ ICA 2026 · 1–4 NOV · IIT ROPAR ◆ ONE LEAF IN · ONE REPORT OUT

A farmer photographs a sick leaf. *What should come back?*

20–40%

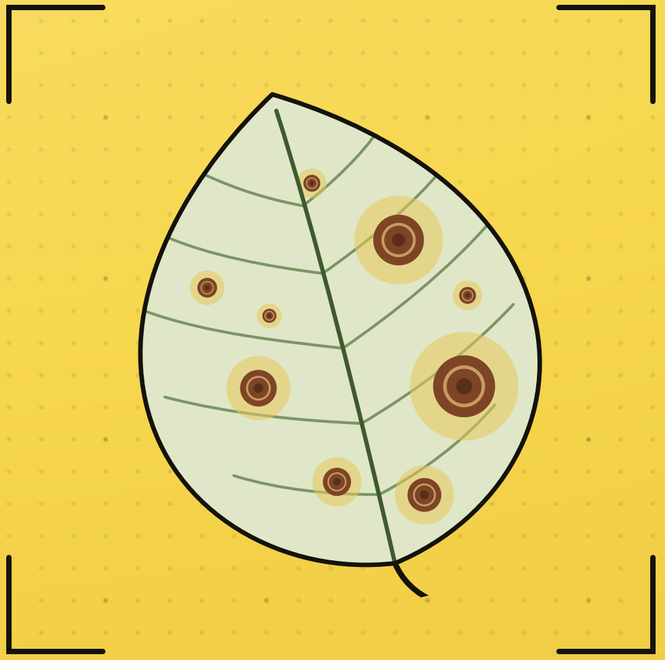
of global crop yield is lost to plant pathogens and pests, every year.

SAVARY ET AL., NATURE ECOL. EVOL. 2019

100s of millions

of smallholder farms. Expert inspection is slow, costly, and does not reach them.

THE BOTTLENECK IS THE AGRONOMIST'S TIME



INPUT · ONE PHOTOGRAPH · ANY RESOLUTION

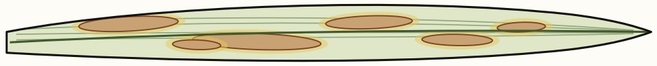
> We asked: can a **small** model read a leaf the way an agronomist writes a report?

KHANNA · SAINI · SINGH · KWATRA

NIT JALANDHAR · ANNAM.AI, IIT ROPAR

A label is not *a diagnosis.*

WHAT A CLASSIFIER RETURNS



“Northern Leaf Blight.”

One label. No severity, no symptoms, no reasoning, **no next step.**

What AgriPerceiver returns

```
"diagnosis": "Northern Corn Leaf Blight",
"type": "fungal",
"severity": 0.62,  "confidence": 0.81,
"symptoms": ["elongated tan lesions, wavy margins",
             "chlorotic halo extending 2-3mm", …],
"reasoning": "Elliptical pale-tan lesions parallel
   to the leaf axis; consistent with
   Exserohilum turcicum under high humidity.",
"recommended_actions": [
   "Apply propiconazole fungicide within 48h", …]
```

CLASSIFICATION

REGRESSION

LANGUAGE

A real model output for a fungal-infected maize leaf (paper, Appendix B). **Classification, regression and language: one decoding objective.**

Why not just ask *a big VLM?*

T^2

Lesions are tiny. *Tokens are expensive.*

Pustules, water-soaking and early chlorosis vanish when an image is resized. Keeping them needs more visual tokens, and attention grows with their **square**.

{ }

Chat models write *prose.*

Decision-support systems consume a fixed **schema**, not paragraphs. Prompting gets you valid JSON. It does not get you the right content.

7–8B

Billions of weights. *One GPU.*

Fully fine-tuning a generalist is out of reach for most agricultural labs, and it risks **forgetting** what the model already knows.

AGRIPERCEIVER, IN THREE LINES

See *sharp.*

5 ANYRES TILES

≈ 2× EFFECTIVE RESOLUTION

Think *small.*

3,645 TOKENS → 128 LATENTS

28.5× COMPRESSION

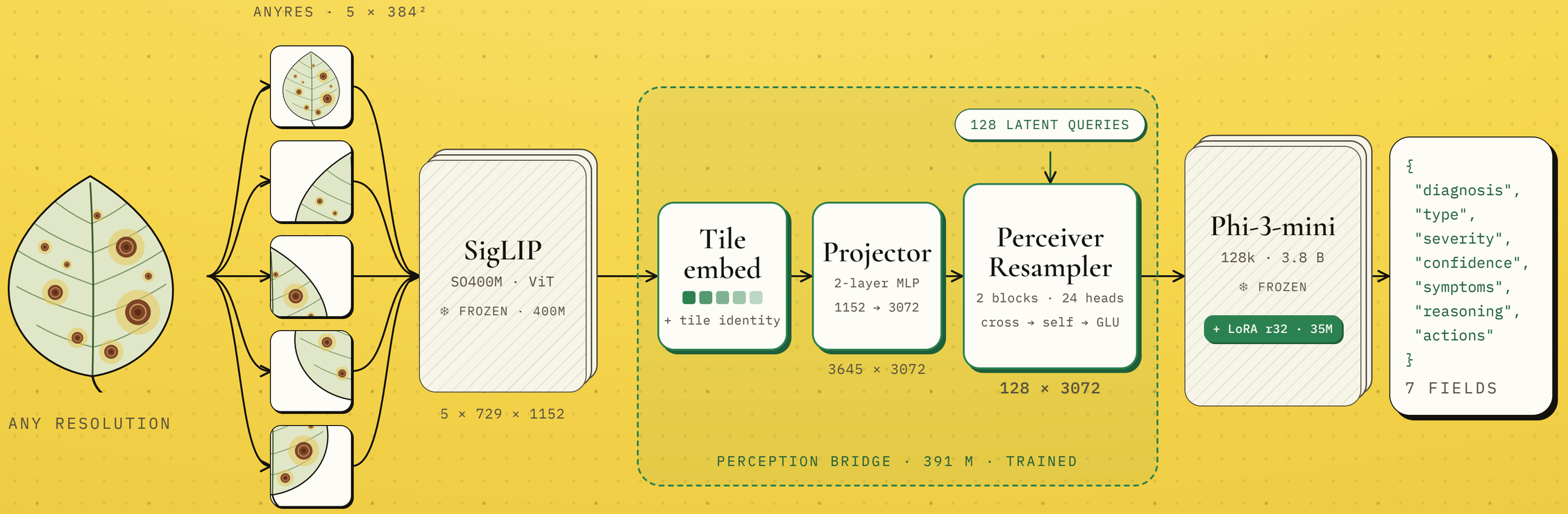
Train *little.*

BOTH BACKBONES FROZEN

9.2% OF 4.65 B PARAMETERS

One leaf in. *One report out.*

FROZEN TRAINED



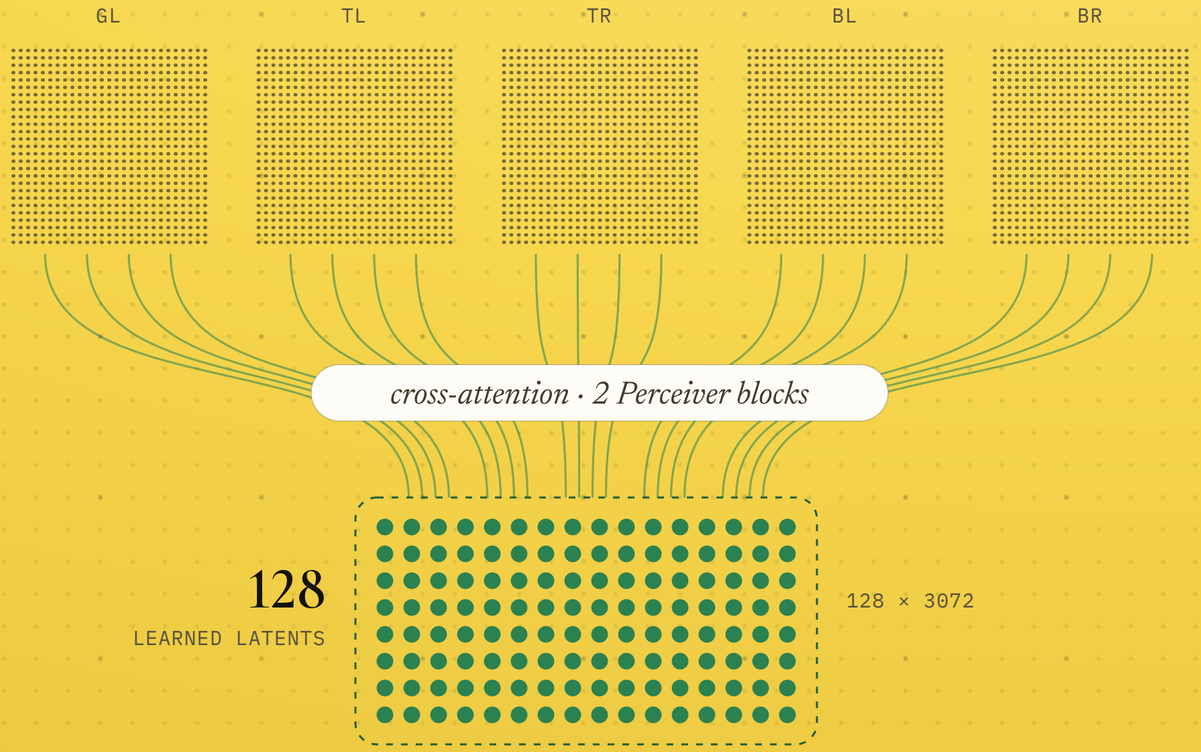
① See sharp. Four quadrants + one global view, encoded by a frozen SigLIP.

② Think small. Tile identity in, 3,645 tokens out as 128 latents.

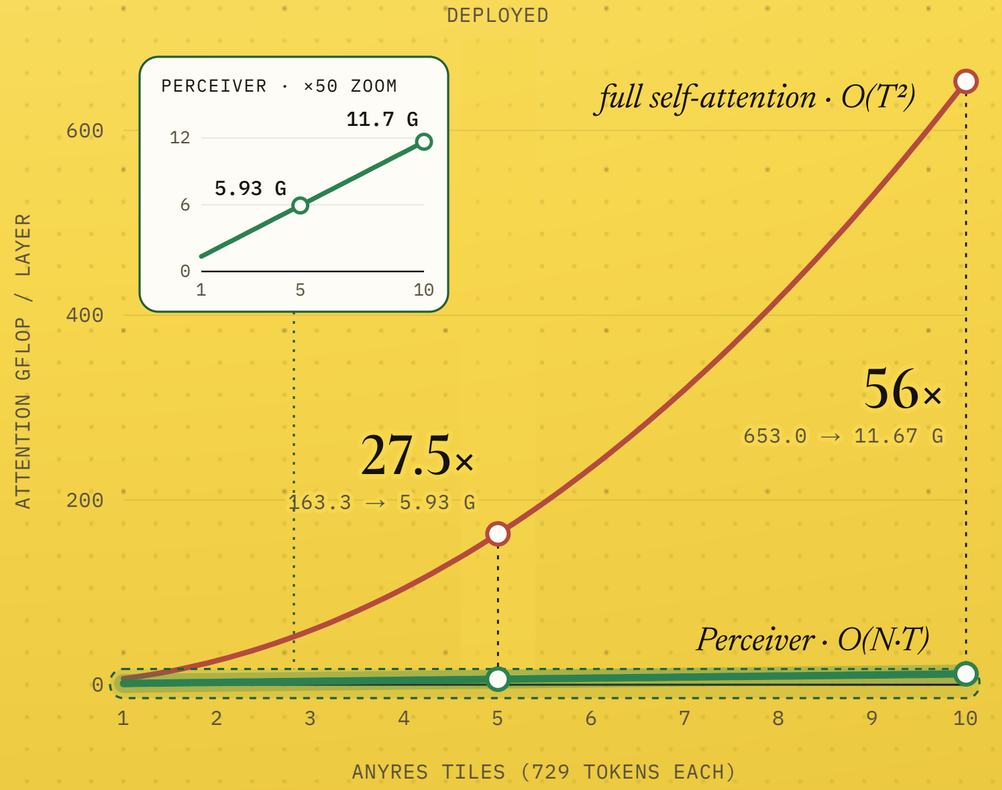
③ Speak in schema. Latents spliced into a frozen LLM.

3,645 tokens in. *128 out.*

3,645 VISUAL TOKENS · 5 TILES × 27 × 27 PATCHES



ATTENTION COST PER LAYER, AS RESOLUTION GROWS



28.5× fewer visual tokens

27.5× fewer attention FLOPs / layer

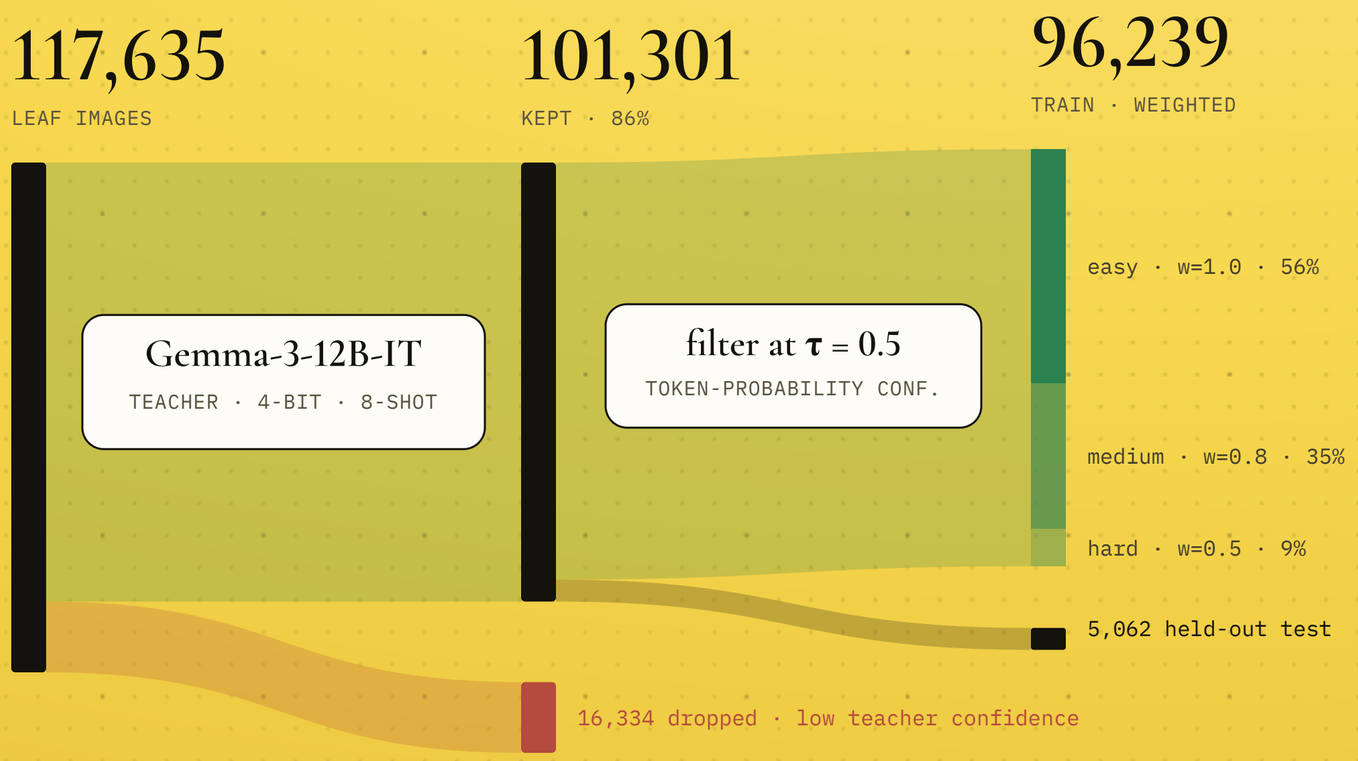
13.29 M → 0.47 M attention entries / head

cost **linear** in resolution

KHANNA · SAINI · SINGH · KWATRA

NIT JALANDHAR · ANNAM.AI, IIT ROPAR

117,635 leaves, *read by a teacher.*



Stage 1 · Align

Train **only the bridge** on image–caption pairs. Both backbones frozen.

391 M · 2 epochs · 14.7 K steps · ≈ 6 h

Stage 2 · Specialise

Bridge + **LoRA** on structured JSON, each sample weighted by the teacher's confidence.

426 M · 9.2% · 3 epochs · 37.9 K steps · ≈ 18 h

ONE NVIDIA H200 · BFLOAT16 · SEED 42

Valid JSON is easy. *The right JSON isn't.*

0.810
composite score
BEST BASELINE 0.615 · +0.195

1.9×
pathology-type macro-F1
0.645 VS 0.336 (INTERNVL2)

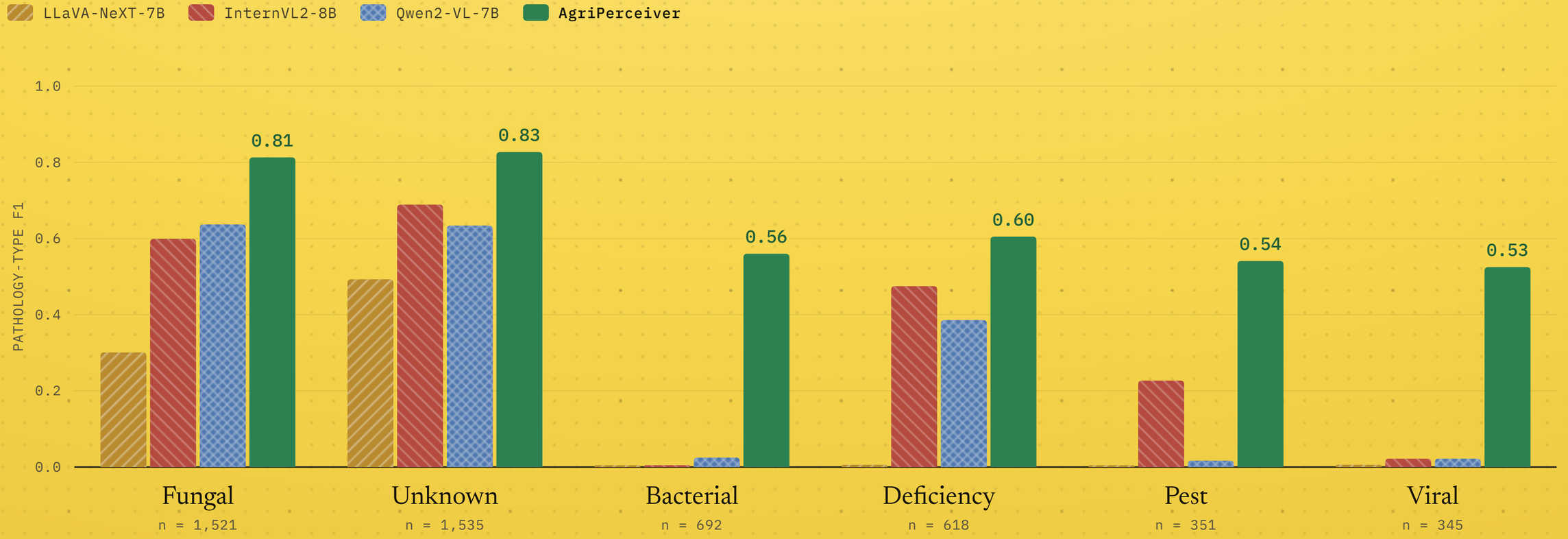
3.3×
lower severity error
MAE 0.067 VS 0.224 (QWEN2-VL)

9.2%
of parameters trained
426 M OF 4.65 B

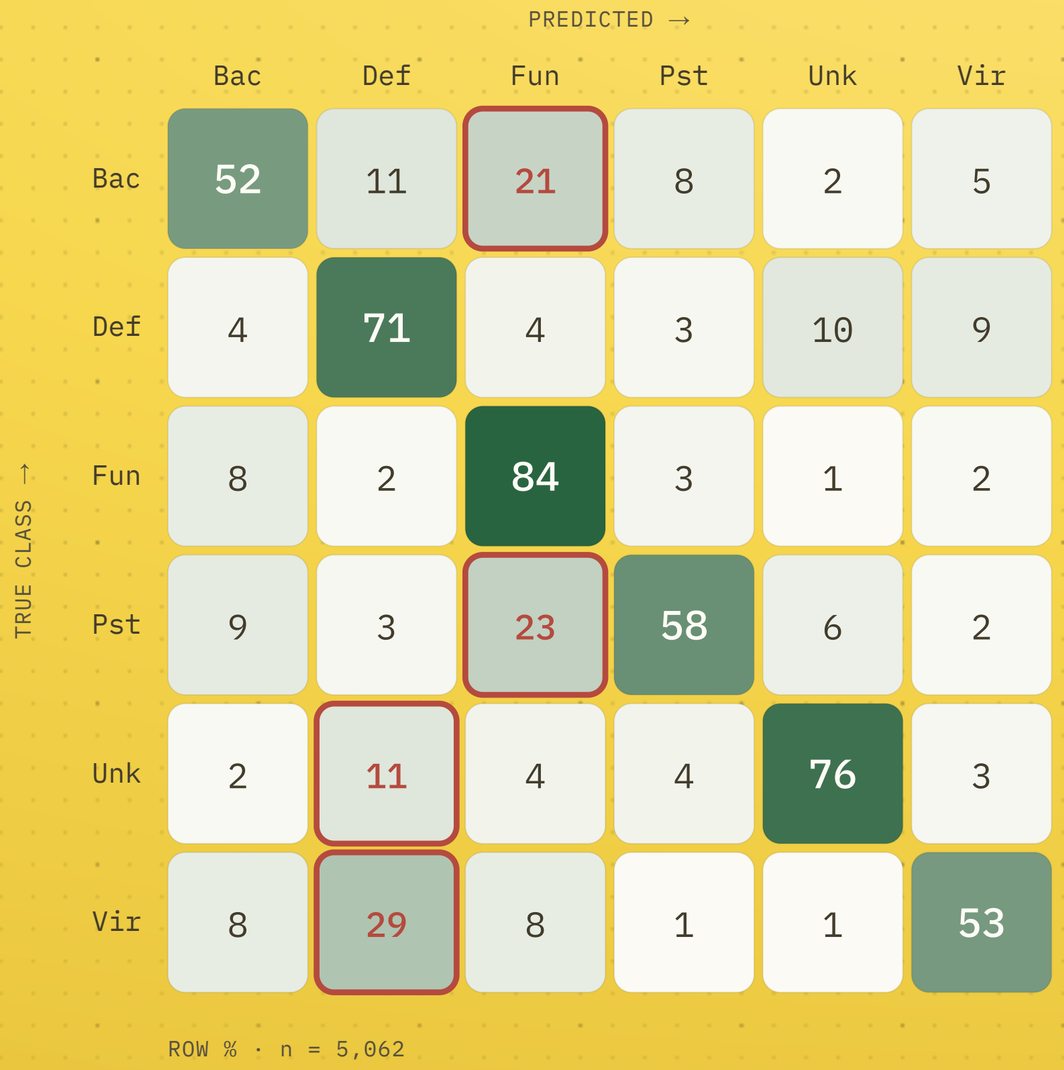
	STRUCTURAL		CLASSIFICATION		REGRESSION		SEMANTIC	
MODEL · 5,062 HELD-OUT LEAVES	JSON %	SCHEMA %	TYPE F1	DIAG. FM	SEV. MAE ↓	SEV. R	BERTSCORE	COMPOSITE
LLaVA-NeXT-7B	99.9	99.9	0.134	0.001	0.344	0.226	0.847	0.590
InternVL2-8B	100.0	100.0	0.336	0.005	0.385	0.624	0.848	0.611
Qwen2-VL-7B	99.9	99.9	0.287	0.001	0.224	0.666	0.830	0.615
AgriPerceiver 4.6 B	**99.7**	**99.7**	**0.645**	**0.486**	**0.067**	**0.855**	**0.921**	**0.810**

◆ STRUCTURE IS NOT THE DIFFERENTIATOR every model emits valid JSON ≥ 99.7% of the time; the gap is entirely in **what the JSON says.**

Generalists see fungus. *They miss the rest.*



On **bacterial** and **viral** leaves every baseline scores F1 ≤ 0.025. AgriPerceiver reaches 0.56 and 0.53, the only model usable on all six classes.



Where it still *gets it wrong.*

29%

Yellow looks like yellow.

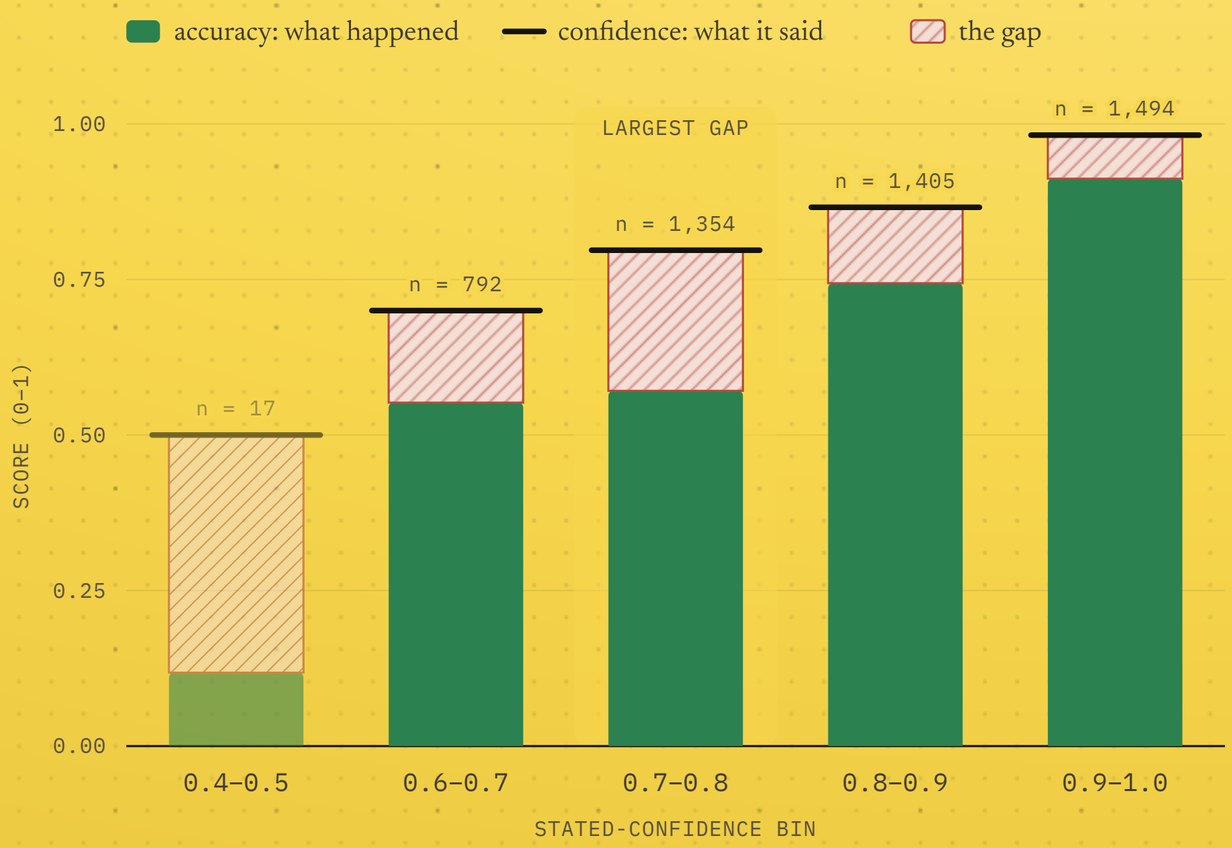
Viral → deficiency. Diffuse **chlorosis** looks the same whether a virus or a missing nutrient caused it. Unknown → deficiency is the largest single cell: 169 leaves.

21–23%

Early lesions look fungal.

Bacterial and pest → fungal. At 384 px per tile, early lesions and fungal **necrosis** overlap.

> These are confusions of *biology*, not of format: clear targets for augmentation.



Only 5 of 10 bins hold predictions: none below 0.4, none between 0.5 and 0.6.

Confident. *A little too confident.*

In the 0.7–0.8 bin, the biggest source of error, the model says **0.80** and is right **0.57** of the time. That gap is inherited: the training targets carry the teacher's own confidence.

EXPECTED CALIBRATION ERROR · LOWER IS BETTER

AgriPerceiver	0.139
InternVL2-8B	0.38
Qwen2-VL-7B	0.37
LLaVA-NeXT-7B	≈ 0*

** not calibration: it answers "unknown" for 98.7% of leaves.*

◆ NEXT: TEMPERATURE SCALING

What this paper does *not* prove. *Yet.*

	SHORTCOMING	WHY IT MATTERS	WHAT CLOSES IT
01	**Our labels come from a model,** not from agronomists.	We measure agreement with Gemma-3-12B. Its mistakes and biases sit inside our ground truth.	*An agronomist-validated gold test set.*
02	**Baselines were prompted,** not fine-tuned.	We show specialisation beats prompting, not yet that our design beats a fine-tuned generalist.	*Fine-tune a baseline on the same 96K labels.*
03	**The spatial parts barely move** the aggregate.	−AnyRes: −0.004 composite. −tile embeddings: ≈ 0. Ablations of latent count and of LoRA need retraining.	*Trained ablations: 64 / 256 latents, no-LoRA.*
04	**Confidence runs high.**	ECE 0.139. A farmer will read 0.8 as 0.8.	*Post-hoc temperature scaling.*
05	**One leaf, one photo,** one moment.	No field trial, no multiple views, no disease progression over time.	*Longitudinal, multi-image tracking.*

The road *from here.*

01 NEXT
Calibrate
Temperature scaling, so that a stated 0.8 means 0.8.

02 NEXT
An expert gold set
Agronomist-validated labels to replace the teacher at test time.

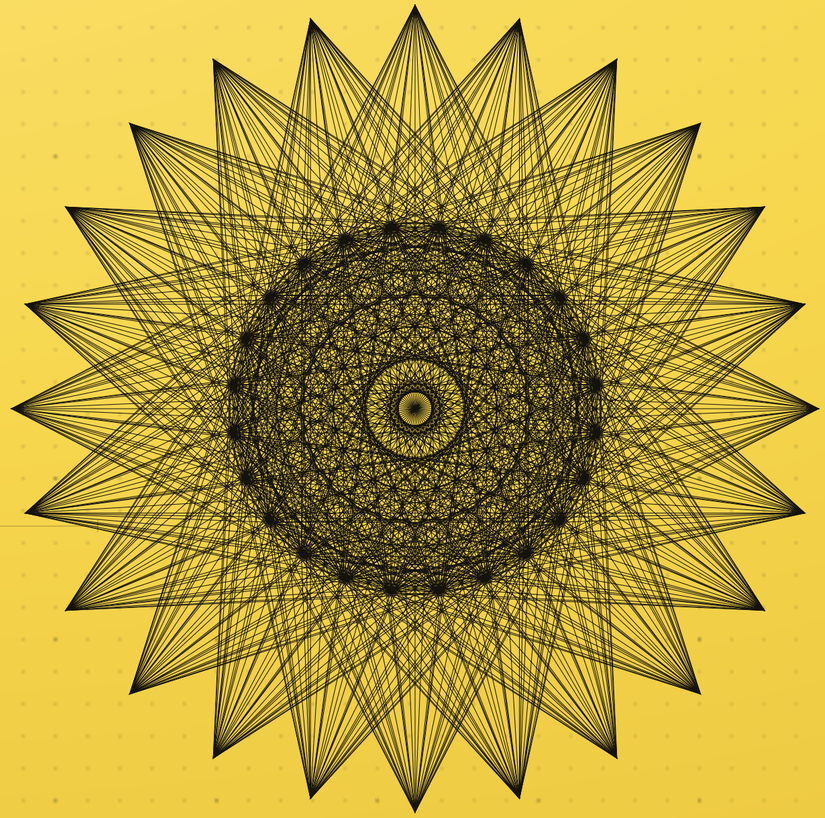
03 SOON
The hard classes
Targeted augmentation for chlorosis and early lesions.

04 SOON
Finish the ablations
Fine-tuned baselines, no-LoRA, 64 / 256 latents.

05 HORIZON
Leaves over time
Longitudinal, multi-image tracking of disease progression.

06 HORIZON
Beyond one crop
Cross-crop transfer, then pilots in real fields beside agronomists.

Every leaf is a report *waiting to be read.*



See sharp.

5 TILES, ≈ 2× RESOLUTION

Think small.

128 LATENTS, 27.5× CHEAPER ATTENTION

Train little.

9.2% OF WEIGHTS → 0.810 COMPOSITE

Thank you.

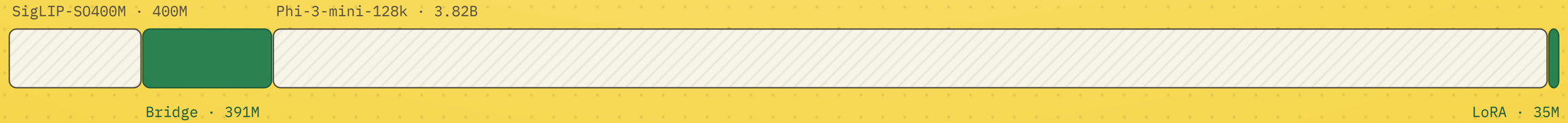
◆ QUESTIONS WELCOME

With gratitude to Dr Roshan Bodile and Dr Rohit Singh (NIT Jalandhar), and to Annam.ai, IIT Ropar, led by Dr Pushpendra Pal Singh, for mentorship and H200 compute.

vatsalk.ec.23@nitj.ac.in · Trying to do our bit in making the world a better place.

Under *the hood.*

WHERE THE 4.65 B PARAMETERS LIVE · GREEN IS TRAINED (426 M · 9.2%)



TRAINING	STAGE 1	STAGE 2
Trainable	bridge · 391 M	bridge + LoRA · 426 M
Optimiser · LR	AdamW · 1e-4	AdamW · 1e-4
Batch (effective)	16	8 × 2 = 16
Epochs · steps	2 · 14.7 K	3 · 37.9 K
Peak VRAM · time	~20 GB · ~6 h	~35 GB · ~18 h
LoRA	none	r 32 · α 64 · drop 0.1

COMPOSITE SCORE · WEIGHTS

0.20 type F1 · **0.15** diag. FM · **0.15** BERT symptoms · **0.10** 1 − MAE · **0.10** BERT reasoning · **0.10** BERT actions · **0.10** JSON valid · **0.05** 1 − ECE · **0.05** schema

Robust to reweighting: doubling any one family keeps a margin of at least +**0.161** over the best baseline.